

Kelp forests

The shallow coastal waters near the shores of many cool oceans support lush beds of seaweed. In some places, such as off Alaska and California, gigantic seaweeds called giant kelp form tall, dense underwater forests that provide food and shelter for a rich variety of marine animals. Some of these animals eat the kelp, but most of them prey on each other.

CLEAR SEAS

On coasts where the water is clear, as it is off the Pacific shores of North and South America, giant kelp—a type of brown seaweed—is able to live in sunlit water up to 130 ft (40 m) deep, and grow as tall as trees on land. Anchored to the seabed, the kelp fronds grow up through the clear water, and their ends trail on the surface. Some fronds can grow to astounding lengths of 165 ft (50 m).

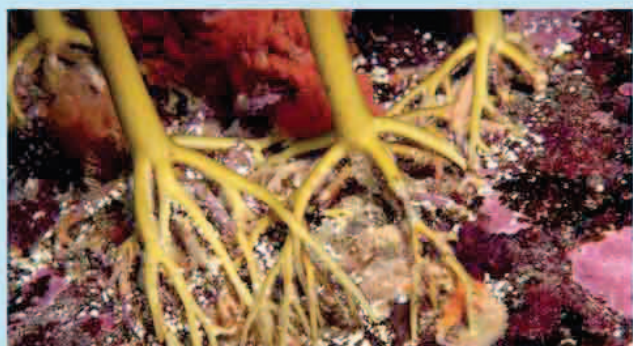
► KELP CANOPY

Buoyed up by gas-filled floats, giant kelp fronds off the coast of California grow straight up toward the sunlight.



STRONG ANCHOR

The kelp attaches itself to rocks on the seabed by a clawlike structure called a holdfast. Although the holdfast may look like the roots of a plant, it does not absorb nutrients in the same way. Its main function is to anchor the kelp to the seabed, so that it is not swept away by the current. The kelp holdfasts are often covered with sponges, barnacles, mussels, and other animals.



WOW!

When the growing conditions are right, giant kelp can grow as much as 24 in (60 cm) each day.